

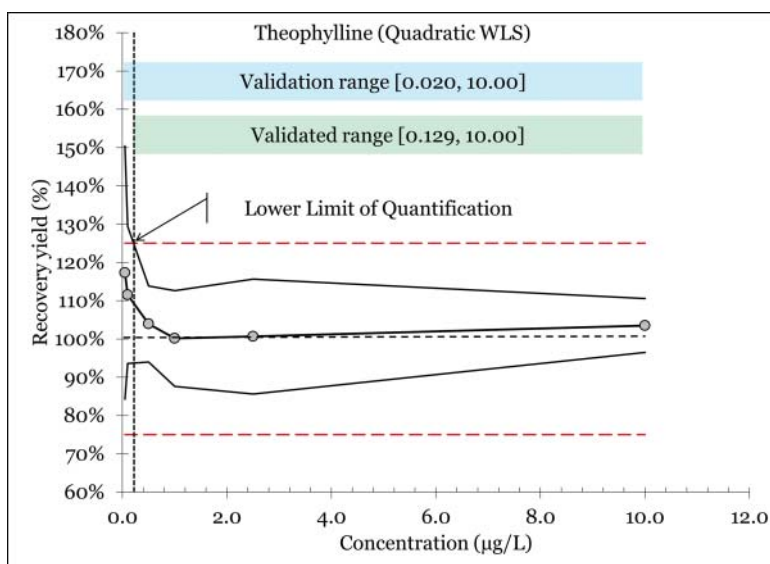


Figure 5.6 THEOPHYLLINE – validation and validated ranges ($\beta\%$ = 80%).

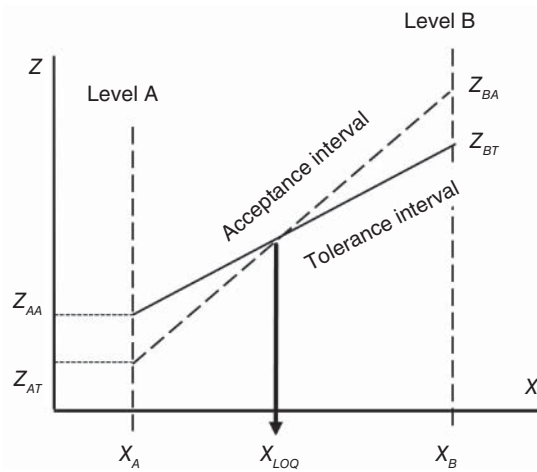
Finally, the concentration where the method is no longer valid is determined by any intersection between one of the β -ETI interpolation segments and the acceptance interval limits. In this example, the crossing happens around 0.1–0.2 $\mu\text{g/l}$. More exactly, it is at 0.129 $\mu\text{g/l}$, and the rationale for accurately calculating this point is explained below. Two concentration ranges can then be defined:

- The validation range *a priori* set by the analyst, i.e. [0.020, 10.00] $\mu\text{g/l}$.
- The validated range is calculated by considering the intersection point between β -ETI and the acceptance interval, when it exists. For this example, the validated range spreads between [0.129, 10.00] $\mu\text{g/l}$. Some FDA regulatory documents also propose to define the two bounds of the validated range as the lower limit of quantification (LLOQ) and the upper limit of quantification (ULOQ), respectively.

An important point when calculating the intersection point to establish the LLOQ is to use the *absolute inverse-predicted concentrations*, as collected in Table 5.3, and no longer the recovery yields, as in Table 5.4. This precaution is compulsory to avoid a bias of the artifact due to the linear interpolation between TI bounds. Proper interpolation curves should be hyperbolas.

Figure 5.7 illustrates the lower part of the accuracy profile, between 0.02 and 0.6 $\mu\text{g/l}$. It shows how to calculate the intersection point. In mathematical terms, it consists in computing the intersection of two straight lines related to absolute inverse-predicted concentrations. This means finding the roots of a system of two equations with two unknowns. This is a typical algebraic problem easily solved with any worksheet. Obviously, when β probability or acceptance interval changes, so is the intersection point modified as well as the bounds of the validated domain.

Figure 5.7 THEOPHYLLINE – lower part of the MAP expressed as absolute inverse-predicted concentration ($\beta\% = 80\%$). The arrow indicates the lower limit of the validated range that can be used as LOQ ($0.129 \mu\text{g/l}$).



To illustrate the method, let us consider the data recorded in Table 5.3 for the THEOPHYLLINE example at levels $A = 0.1 \mu\text{g/l}$ and $B = 0.5 \mu\text{g/l}$, since it is between these two levels that one segment of the tolerance interval profile crosses of the upper limit of acceptability. As stated, it is compulsory to use absolute values. The following notations reported in Figure 5.8 will help to understand the computation:

- Z recovered concentration values on the Y -axis.
- X known concentration values on the X -axis
- X_A and X_B concentrations of the selected levels A and B , respectively.
- Z_{AT} and Z_{BT} tolerance interval upper bounds at selected levels.
- Z_{AA} and Z_{BA} acceptance interval upper bounds at selected levels.
- X_{LOQ} concentration at the crossing point defined as the LOQ.

These two straight lines can be represented by a system of two equations, the first describes the acceptance limit, which varies with the concentration as it is defined as a percentage, and the second the tolerance interval interpolation segment. Since, by construction, the acceptance interval always passes through zero, we must find $A_0 = 0$.

Equation (1.1) for the acceptance interval	$Z = A_0 + A_1 X$
Equation (1.2) for the tolerance interval	$Z = T_0 + T_1 X$

The slope and intercept of Eq. (1.1) are calculated as follows:

Slope of Eq. (1.1)	$A_1 = \frac{Z_{BA} - Z_{AA}}{X_B - X_A}$
Intercept of Eq. (1.1)	$A_0 = Z_{AA} - A_1 X_A$
Symmetrically for Eq. (1.2):	
Slope of Eq. (1.2)	$T_1 = \frac{Z_{BT} - Z_{AT}}{X_B - X_A}$
Intercept of Eq. (1.2)	$T_0 = Z_{AT} - T_1 X_A$

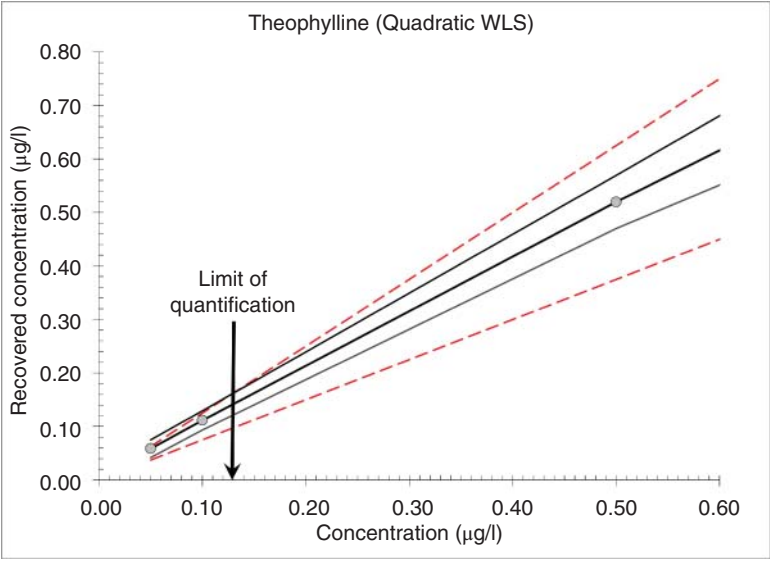


Figure 5.8 Schematic representation for LOQ calculation.

Finally, the abscissa of the crossing point is given by:

$$X_{LOQ} = \frac{A_0 - T_0}{T_1 - A_1}$$

To illustrate the calculation, a worksheet can easily be implemented as illustrated. Finally, $X_{LOQ} = 0.129 \mu\text{g/l}$. It must also be noted that $A_0 = 0$ as expected. When the proportion β is modified, so is the LOQ. For instance, when $\beta\% = 67\%$, it becomes $X_{LOQ} = 0.099 \mu\text{g/l}$. This remark confirms the usefulness of this proposal to define the LOQ, but harmonization is necessary to make it acceptable. Chapter 9 suggests a more extensive discussion about this fundamental validation parameter.

	A	B	C	D
1	Limit of Quantification from the Accuray Profile			
2	Levels	A	B	
3	Concentration (µg/L)	0.100	0.500	
4	Upper bounds Tolerance interval	0.125	0.625	
5	Upper acceptance limit	0.129	0.569	
6	Equation coefficients			
7	Tolerance interval	Slope	1.250	= $(C4-B4)/(C3-B3)$
8		Intercept	0.000	= $B4-C7*B3$
9	Acceptance interval	Slope	1.099	= $(C5-B5)/(C3-B3)$
10		Intercept	0.019	= $B5-C9*B3$
11	Limit Of Quantification		0.129	= $(C10-C8)/(C7-C9)$

5.3 Statistical Dispersion Intervals

An interval for Z is a set of real numbers for which $a \leq Z \leq b$, where a and b are real numbers called the bounds or the limits of the interval. It is used together with the symbol $[a, b]$. Globally there are two categories of statistical intervals, one is used to describe, the other to predict.

For instance, confidence intervals (CI) are the best-known, widely explained and used in the statistical literature. It is a descriptive interval around the computed value of a parameter, such as a mean or a standard deviation. It is assumed to contain the theoretical *true* value of the corresponding parameter with a certain probability of error conventionally denoted α . True value is likely to lie in this interval; the adjective “*likely*” relates to the confidence level of the confidence interval, classically denoted $1 - \alpha$. The practical definition of the CI is available in ISO 23833:2013: “range of analytical error expected to contain the true value with a stated uncertainty as estimated from a statistical model of the measurement process.” Any true statistical parameter of a population is assumed to be bounded by the confidence interval.

In contrast, there are at least two other categories of statistical dispersion intervals that are less well known but extensively discussed here, namely the prediction intervals and the tolerance intervals (TI). In the prospect of method validation and measurement uncertainty, they are particularly interesting. Unfortunately, the definitions are confusing.

In statistical literature, there are references to both tolerance intervals, sometimes referred to as:

- β - γ -content tolerance interval (β - γ -CTI).
- β -expectation tolerance interval (β -ETI).

TIs were introduced by the end of the 1940s when quality assurance and statistical process control principles became widespread in industry. Table 5.5 proposes a classification of different statistical intervals noted from A to D, depending on the study’s goal. It is compiled and adapted from [24].

The *prediction interval* is an interval that will “contain a future randomly selected observation from a distribution.” With a specified degree of confidence. It is generally useful to predict the result of one, or a small number, of future measurements. Prediction intervals for all future observations are of interest only if a small number of measurements are produced because they are often very wide. Also, the exact number of future measurements is sometimes not known or may conceptually be infinite. Moreover, rather than requiring that the calculated interval contain a specified number of units, it is generally sufficient to construct an interval to contain a substantial proportion of such units.

Table 5.5 Examples of some statistical intervals.

Goal	Description	Prediction
Range	A. Tolerance interval to contain (or cover) at least a specified proportion of a distribution	B. Prediction interval to contain the observations from a future sample
Probability	C. Confidence interval for the probability of an observation being less than (or greater than) some specified value	D. Prediction interval to contain the proportion of observations in a future sample that exceed a specified limit

Source: Adapted from Meeker et al. [24].

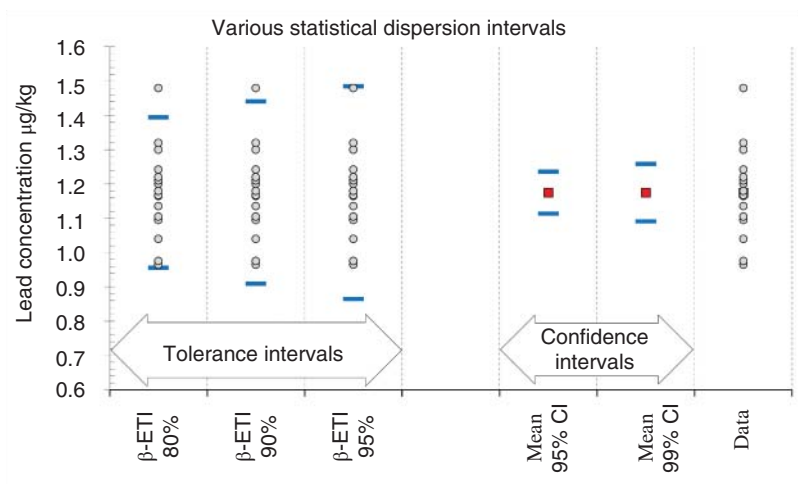


Figure 5.9 Comparison tolerance and confidence intervals calculated for 18 replicates assumed to be normally distributed.

Specifically, the TI is an interval that one can claim to “contain at least a specified proportion, β , of the distribution” with a specified degree of confidence usually denoted γ in this context but corresponding to the classic level of confidence used for CI. Such an interval is of particular interest in limiting the process capability for measurements produced in massive quantities. This contrasts with a prediction interval which is, as defined, of greatest interest in predicting a small number of future units.

Thus, it was demonstrated that a β -expectation tolerance interval ($\beta\text{-ETI}$) can also be defined as a prediction interval for a single future observation.

To illustrate how these distinct types of statistical intervals may apply, Figure 5.9 compares the intervals obtained with 18 replicates performed on a single sample and assumed to be normally distributed¹. This is the simplest situation and formulas used for this example are not presented but can be found in many publications, such as [25]. The major difference between tolerance and confidence intervals appears: the TI is applied to describe the whole data distribution, while the CI is used to characterize a statistical parameter (the mean for this example) which is a combination of data.

Apparently, there is some confusion among the definitions of the term *tolerance* as used in standards or guidelines. In ISO 16269-6 standard, a TI is “statistical dispersion interval,” and in ISO 12669-8 standard, it is a “prediction interval.” Moreover, tolerance can also be defined as a complement to acceptance. For instance, in ISO/IEC Guide 98-4, the following definition is proposed for the tolerance interval “the interval of permissible values of a property” and, at the opposite, the rejection interval is “the interval of non-permissible measured values.” In the same document, the acceptance interval is “the interval of permissible measured quantity values” [26]. The concepts are reviewed in Section 8.2 on the conformity assessment of a

¹ Unpublished personal data